
AN0005-EC NB-IoT LwM2M Application Note_V1.3

EigenCOMM Wireless Microcontroller

Document Description

the purpose of present document is to help users quickly connect their module with LwM2M server through AT commands.

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1. Foreword

1.1 Scope

The present document describes how to use LwM2M AT command to implement user's module logging in to public LwM2M server, and implement module monitoring from the server.

1.2 LwM2M protocol

LwM2M is a protocol applicable to the Internet of Things. The full name is Lightweight Machine-To-Machine, proposed and defined by OMA (Open Mobile Alliance). The main entities of the protocol include LwM2M Server, LwM2M Client and LwM2M Bootstrap Server. LwM2M Server as the server, deployed at the service provider, LwM2M Client as the client, deployed on each LwM2M device. In addition, you can also deploy LwM2M Bootstrap Server to guide clients as needed. The basic architecture of LwM2M is shown below:

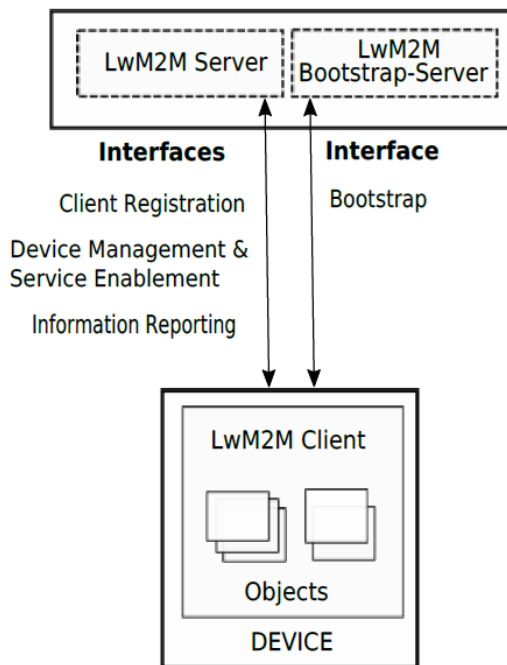


Figure 1. The overall architecture of the LwM2M Enabler

1. Bootstrap Interface

The Bootstrap Interface is used to provision essential information into the LwM2M Client to enable the LwM2M Client to perform the operation “Register” with one or more LwM2M Servers.

2. Client Registration Interface

The Client Registration Interface¹ is used by a LwM2M Client to register with one or more LwM2M Servers, maintain each registration and de-register from a LwM2M Server.

3. Device Management and Service Enablement Interface

The Device Management and Service Enable Interface is used by the LwM2M Server to access Object Instances

and Resources available from the LWM2M Client. The interface provides this access with the use of “Create”, “Read”, “Write”, “Delete”, “Execute”, “Write Attributes”, or “Discover” operations.

4. Information Reporting Interface

The Information Reporting Interface is used by a LWM2M Server to observe any changes in a Resource on a LWM2M Client, receiving notifications when new values are available.

LWM2M protocol is based on RESTFUL architecture. The LWM2M Enabler uses the Constrained Application Protocol (CoAP) with UDP. Datagram Transport Layer Security (DTLS) provides security for UDP transport layer. LWM2M defines a model with resources as the basic unit, and each resource represents one of the available information in the LwM2M client. Resources all exist in Object Instance, which is the instantiation of Object. The protocol allows users to customize objects to describe their products according to actual needs. For more detailed description of the LwM2M, please visit <https://www.omaspecworks.org>

1.3 Brief introduction

The present document describes how to use the LwM2M server provided by Eclipse. This server is mainly used by LwM2M developers for testing, only provides some basic functions that follow the standard LwM2M protocol. When the customer's product is finally commercialize, it is necessary to develop an LwM2M server that meets its own needs, or use other commercial LwM2M servers.

The present document also introduces the registration process of the LwM2M client, the operation flow of various instructions, and the detailed explanation of the relevant AT instruction parameters. Finally, a few suggestions for low power consumption are provide.

2. LwM2M Server

The common LwM2M server address is coap: //leshan.eclipseprojects.io. It is a public LwM2M server provided by Eclipse. It provides two access methods: encrypted (port number 5684) and unencrypted (port number 5683). Allows users to test their LwM2M client. In addition, an https server is provide: <https://leshan.eclipseprojects.io>, users can operate their LwM2M client on the webpage.

For backup, the present document also provides an LwM2M server for testing:

coap: //Leshan.eigencomm.com. It also provides an http server <http://leshan.eigencomm.com>. The present document's examples are all connected to this server.

2.1 Non-encrypted client

To non-encrypted client, no configuration is required on the server, just register directly.

2.2 Encrypted client

If the client wants to connect to the server through DTLS, you need to click the SECURITY page on the webpage to add security configuration for your client, as shown in the figure below.

Note: the present document does not fully test encrypted clients, so the examples listed are non-encrypted clients

LESHAN CLIENTS **SECURITY**

The Leshan Certificate (x509v3 der encoded)

Hex:

```
308201863082012CA0030201020209008307C59CC28607E6300A06082A8648CE3D04
0302301D311B301906035504030C124C657368616E205365727665722044656D6F30
20170D313830393230313632303235A180F3231313830383237313632303235A30
1D311B301906035504030C124C657368616E205365727665722044656D6F30593013
06072A8648CE3D020106082A8648CE3D030107034200048BD50F73E21D8FA3042C83
D21E85570ECDEEF0C1149BE054FC9263FAB6D430BF8B9C918746FA1897192B28F2F
2AF2A1DEEDF2818DE4C276C315FF70D4A570D8A3533051301D0603551D0E04160414
4D0B724424C7534752BCDD448C7B2D8CE045F5B301F0603551D230418301680144D
0B724424C7534752BCDD448C7B2D8CE045F5B301F0603551D130101FF0405300301
01FF300A06082A8648CE3D040302034800304502202F211231D8590AC223383726D0
D5BCF2D0382E38260A3C6778F5C06983024D90022100BF05DEA8E68A29F3D866CBF
5F320ED05EB69FF07267637F4522DF1F8E640D3E
```

Base64:

```
MIIBhjCCASygAwIBAgIJAIHhZzChgfmMAoGCCqGSM49BAM
CMB0xGzAZBgNVBAMMEkxlc2hhbiBTZXJ2ZXIgdGVtbzAgFw
0xODA5MjA4MjIwMTUyMjA0MjIwMTUyMjA0MjIwMTUyMjA0
BkGA1UEAwVSTGVzaGFuIFNlcnZlcjE8ZWN1bWVkaWYwYHkx
zj0CAQYIKoZIzj0DAQcDQgAEi9UPc+Idj6MELIPSHoVXD3s
u8MEUM+sFT8kmP6ttQvv4uckYdG+hIXG5so8vKvKh3u3ygY
3kwnbDFf9w1KV9iKNtMFwHQYDVR0BBYEF0LckQkx1NHU
svN1EjHstj0BF9bMA8GA1UdIwQYMBaAF0LckQkx1NHUsvN
1EjHstj0BF9bMA8GA1UdEwEB/wQFMAMBaf8wCgYIKoZIzj0
EAWIDSAAwRQIgLvESMdtZCsIjODcm0N88tA4LjgmCjxneP
XAaYmCTZACIQ/Bd5qjmiinz2GbL9fmg7QXraF8HJvY39FI
t8fjmQNPg==
```

Clients generally need it for X509 authentication.

The Leshan Public Key (SubjectPublicKeyInfo der encoded)

Elliptic Curve parameters: secp256r1 [NIST P-256, X9.62 prime256v1] (1.2.840.10045.3.1.7)

Public x coord: 8bd50f73e21d8fa3042c83d21e85570ecdeef0c1149beb054fc9263fab6d430b

Public y coord: f8b9c918746fa1897192b28f2af2a1deedf2818de4c276c315ff70d4a57d88

Hex:

```
3059301306072A8648CE3D020106082A8648CE3D030107034200048BD50F73E21D8F
A3042C83D21E85570ECDEEF0C1149BE054FC9263FAB6D430BF8B9C918746FA18971
92B28F2AF2A1DEEDF2818DE4C276C315FF70D4A57D88
```

Base64:

```
MFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAEi9UPc+Idj6M
ELIPSHoVXD3s3u8MEUM+sFT8kmP6ttQvv4uckYdG+hIXG5so
8vKvKh3u3ygY3kwnbDFf9w1KV9iA==
```

Clients generally need it for RPK authentication.

Add new client security configuration

Figure 2: Security configuration

For example, if the user wants to use the pre-shared key encryption method, click add new client security configuration in the figure above to add it. Identity can be set to the name of your product. Key can set to the IMEI number of the device.

New security configuration

Client endpoint:

Security mode:

Identity:

Key:

Hexadecimal format

Figure 3. New security configuration

After that, you can see in the list:

ec-client-116	Pre-Shared Key	Identity : eigencomm Key : 201901	
---------------	----------------	--	---

Figure 4. Security configuration

3. LwM2M client

3.1 Object selection

The current standard object definitions are mainly defined by the IPSO Alliance and OMA, see www.omaspecworks.org. If you use a non-standard custom object id, you can only display the id and instance of the object when you first enter the web page of the LwM2M server used in this article. After reading the entire instance, you can display the resource id of the object, and then you can use the resource id. Do it. To customize a new object, visit <http://www.openmobilealliance.org/wp/OMNA/LwM2M/LwM2MRegistry.html>

3.2 Client online

The client in the present document are non-encrypted client. The present document choose the AT command.

The steps for going online are as follows:

1. After powering on the module, enter ATQ0 from the serial port. Ensure the active reporting messages can be output from the serial port.
2. Create an LwM2M client instance and register with the LwM2M server.

AT+LWM2MCREATE="Leshan.eigencomm.com",5683,56830,"ec-lient-128",86400

If successful, the serial port will output

+LWM2MCREATE: 0

OK

If failure, returns

+LWM2M ERROR: XXX

For detailed description of AT command, please refer to the document "*EiGENCOMM EC NB-IoT AT Command Manual.docx*" or section 3.4.

After the AT command is successfully executed, the client page of the webpage can see the device:

ec-client-128	yMjffidQO8	Jan 28, 2019 2:23:02 PM	Jan 28, 2019 2:24:36 PM	
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Figure 5. device online

3. Add object id:3303

The device has two default objects, namely the LwM2M Server with id 1 and Device with id 3. They are standard attributes defined by OMA. The current attribute values are default values, as shown in the figure below. If the customer wants to use these two standard objects, can make a request.

LwM2M Server /1		
Instance 0	/1/0	Create New Instance Observe ▶ Read Write Delete
Short Server ID	/1/0/0	123
Lifetime	/1/0/1	60
Default Minimum Period	/1/0/2	0
Default Maximum Period	/1/0/3	0
Disable	/1/0/4	
Disable Timeout	/1/0/5	0
Notification Storing When Disabled or Offline	/1/0/6	false
Binding	/1/0/7	U
Registration Update Trigger	/1/0/8	

Device /3		
Instance 0	/3/0	Observe ▶ Read Write Delete
Manufacturer	/3/0/0	Open Mobile Alliance
Model Number	/3/0/1	Lightweight M2M Client
Serial Number	/3/0/2	345000123
Firmware Version	/3/0/3	1.0
Reboot	/3/0/4	
Factory Reset	/3/0/5	
Available Power Sources	/3/0/6	0=1, 1=5
Power Source Voltage	/3/0/7	0=3800, 1=5000
Power Source Current	/3/0/8	0=125, 1=900
Battery Level	/3/0/9	100
Memory Free	/3/0/10	15
Error Code	/3/0/11	0=0
Reset Error Code	/3/0/12	
Current Time	/3/0/13	2013-05-02T18:47:33+08:00
UTC Offset	/3/0/14	+01:00
Timezone	/3/0/15	Europe/Berlin
Supported Binding and Modes	/3/0/16	U
Device Type	/3/0/17	
Hardware Version	/3/0/18	
Software Version	/3/0/19	
Battery Status	/3/0/20	
Memory Total	/3/0/21	
ExtDevInfo	/3/0/22	

Figure 6. default object

Add device object. Take temperature sensor 3303 as an example. The device has four attributes: 5601, 5602, 5700, and 5701.

AT + LWM2MADDOBJ = 0,3303,0,4, "5601; 5602; 5700; 5701"

Temperature /3303		
Instance 0	/3303/0	Create New Instance Observe ▶ Read Write Delete
Min Measured Value	/3303/0/5601	
Max Measured Value	/3303/0/5602	
Min Range Value	/3303/0/5603	
Max Range Value	/3303/0/5604	
Reset Min and Max Measured Values	/3303/0/5605	
Sensor Value	/3303/0/5700	
Sensor Units	/3303/0/5701	

Figure 7. add object

After refreshing the page, you can see the added objects. Although only 4 attribute values are add to the client, the server still displays all the attribute values of this object defined by OMA.

3.3 Instruction operation process

3.3.1 Read/write/execution instructions

Click to read attributes 5601 from the server. The server will send a read instruction to the device. Module will output the unsolicited AT command: +LWM2MREAD: 0,3303,0,5601. At present, the command of the Leshan server has a timeout of 5 seconds. If no response receive within 5 seconds, the timeout is prompted. The module should send a response to the read command, for example: AT +LWM2MREADCONF = 0,3303,0,5601,3,4, "3.14". (For a detailed description of AT, see “*EiGENCOMM EC NB-IoT AT Command Manual.docx*” or section 3.4).

After the server receives the response, it will display on the page.

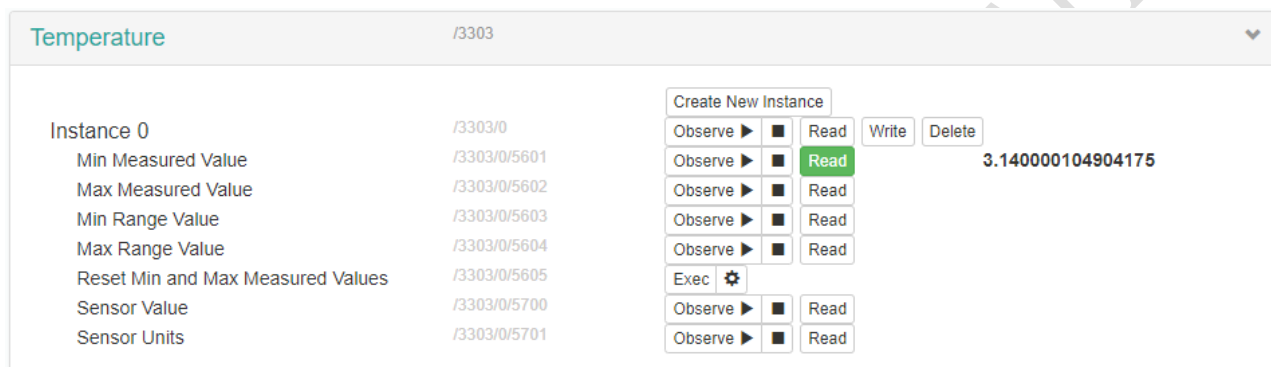


Figure 8. read resource

The write instruction and execution instruction are similar to the read instruction. The command flow is shown in the figure below.

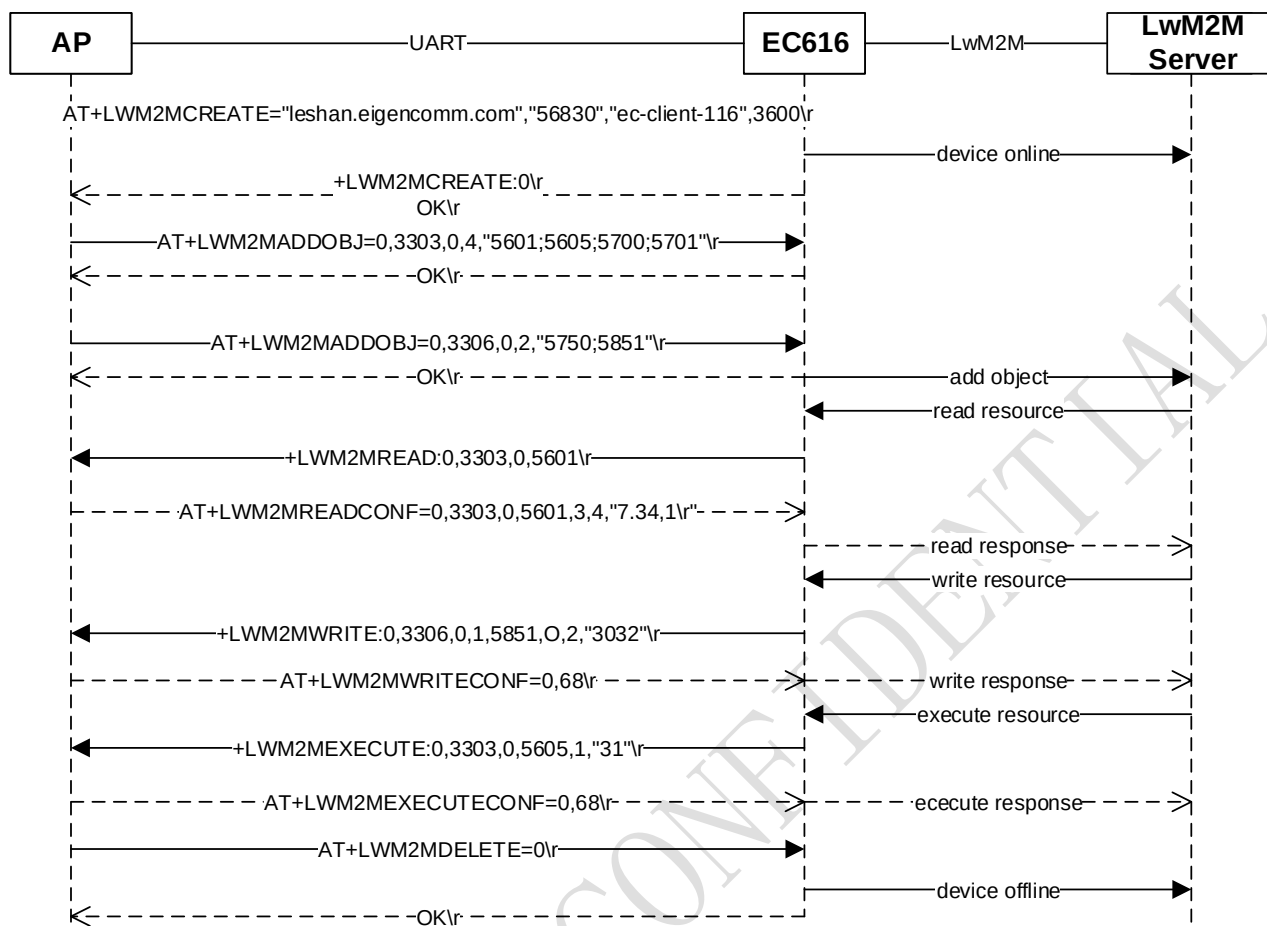


Figure 9. AT command flow

3.3.2 Observe and Notify instructions

Click the observe command on the web page, as shown in the following figure, indicating that the server wants to observe object: 5601. In the design of the lwm2m protocol, various thresholds can be set for the resources of the observed object. When the resource threshold is reach, the client reports the value of the resource. If the resource is observed, the client will continuously read the resource value to determine whether the resource has reached the threshold. At present, the https server of Leshan does not have the function of setting thresholds for resources, and because the present document adopts the AT method to interact with MPU, considering the low power consumption requirements, frequent reading of MPU will increase power consumption. Therefore, the present document adopts MPU to send a notify message via AT command to inform the client that the value of a certain resource has changed, and then the client sends an unsolicited AT command back to read the resource. After the MPU responds, it notifies the server to display. Like: AT+LWM2MNOTIFY:0,3306,0,5601

The present document provide another method to send the value directly from client to server. Like: AT+LWM2MNOTIFY=0,3303,0,5601,3,4,"3.14"

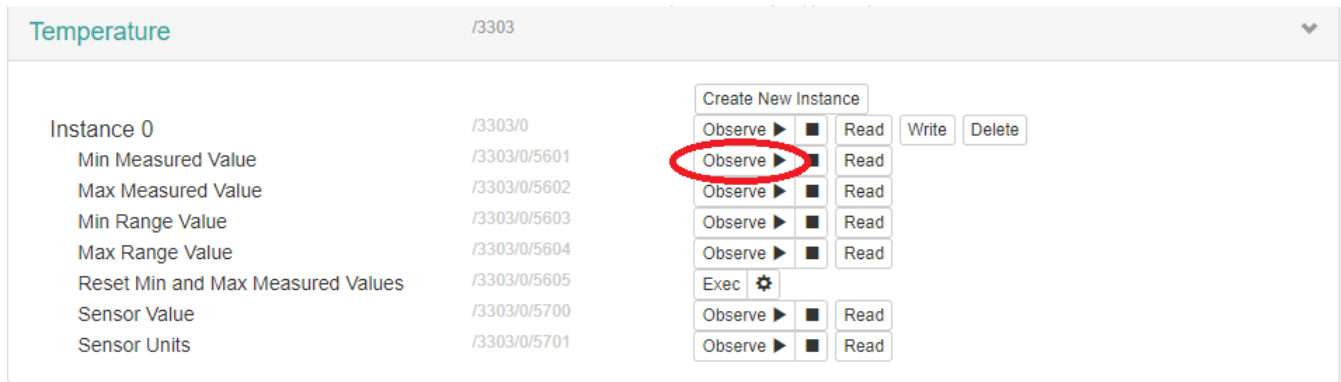


Figure 10. observe resource

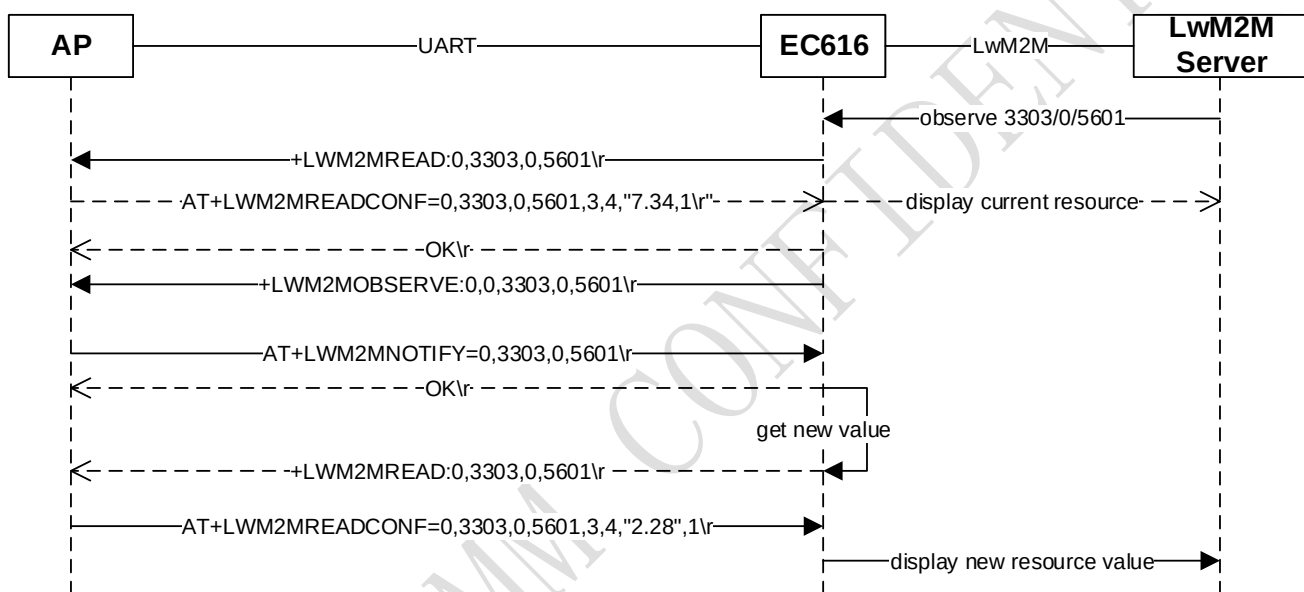


Figure 11. observe and notify AT flow

3.4 LwM2M Commands

3.4.1 AT+LWM2MCREATE

This command creates an instance of lwM2M client and register with lwM2M server. It need specify sever, port, ender point name, lifetime. Also can set psk id and psk if need DTLS.

AT+LWM2MCREATE

Set Command AT+LWM2MCREATE=<server>,<port>,<local port>,<endpoint>,<life time>[,<psk id>,<psk>]	Response +LWM2MCREATE:<clientId> OK If there is any error, response: + LWM2M ERROR:<err>
Test Command AT+LWM2MCREATE=?	Response +LWM2MCREATE: "<server >",<range of

	supported<port>),(range of supported<local port>), "<endpoint>",(range of supported <lifetime>),"<psk_id>","<psk>" OK
Maximum Response Time	65s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type LwM2M client's id
<server>	String type LwM2M server's URL or IP address
<port>	Integer type LwM2M server's port number
<local port>	Integer type LwM2M client's local port
<endpoint>	String type LwM2M client's endpoint name
<life time>	Integer type LwM2M client's life time
<psk id>	String type LwM2M client's public identity
<psk>	String type LwM2M client's pre shared key

Example:

AT+LWM2MCREATE="leshan.eigencomm.com",5683,56830,"client0",86400

+LWM2MCREATE: 0

OK

AT+LWM2MCREATE="leshan.eigencomm.com",5684,56840,"ec-client-116",86400,eigencomm,323031393031

+LWM2MCREATE: 0

OK

Note:

The parameter server is URL or IP address, no need to add a prefix like "coap://"

If use the common LwM2M server coap: //leshan.eclipseprojects.io. the AT command is:

AT+LWM2MCREATE="leshan.eclipseprojects.io",5683,56830,"client0",86400

+LWM2MCREATE: 0

OK

This is the only difference.

If using an encrypted client, make sure that the server has been configured with client security configuration, and matches the client's psk id/psk, and the port must be 5684

3.4.2 AT+LWM2MDELETE

This command deletes a specified lwM2M client instance.

AT+LWM2MDELETE

Set Command AT+LWM2MDELETE=<clientId>	Response OK If there is any error, response: + LWM2M ERROR:<err>
Test Command AT+LWM2MDELETE=?	Response +LWM2MDELETE:(list of supported<clientId>) OK
Maximum Response Time	65s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
------------	--

3.4.3 AT+LWM2MADDOBJ

This command adds a lwM2M object to a specified lwM2M client instance. Definitions of object, instance and resource. Please refer to Lightweight Machine to Machine Technical Specification, ext-label Objects Produced by IPSO Alliance and oma-label Objects Produced by OMA.

<http://www.openmobilealliance.org/wp/OMNA/LwM2M/LwM2MRegistry.html>

AT+LWM2MADDOBJ

Set Command AT+LWM2MADDOBJ=<clientId>,<objectId>,<instanceId>,<resourceCount>,<resourceIds>	Response OK If there is any error, response: +LWM2M ERROR:<err>
Test Command AT+LWM2MADDOBJ=?	Response +LWM2MADDOBJ:(list of supported<clientId>),(range of supported<objectId>),(range of supported<instanceId>),(range of supported<resourceCount>), "<resourceIds>" OK
Maximum Response Time	65s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type
------------	--------------

	LwM2M client's id returned by +LWM2MCREATE
<objectId>	Integer type Object id number
<instanceId>	Integer type Instance id number
<resourceCount>	Integer type Number of resources
<resourceIds>	String type Resources numbers separated by semicolons

Example:

AT+LWM2MADDOBJ:0,3306,111,3,"5750;5850;5851"

OK

3.4.4 AT+LWM2MDELOBJ

This command deletes an object from a specified LwM2M client instance.

AT+LWM2MDELOBJ

Set Command AT+LWM2MDELOBJ=<clientId>,<objectId>	Response OK If there is any error, response: +LWM2M ERROR:<err>
Test Command AT+LWM2MDELOBJ=?	Response +LWM2MDELOBJ:(list of supported<clientId>),(range of supported<objectId>) OK
Maximum Response Time	65s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<objectId>	Integer type Object id number

3.4.5 +LWM2MREAD Indicator of LwM2M server's reading command

This is an unsolicited message to represent LwM2M client receive the LwM2M server's reading command.

+LWM2MREAD

+LWM2MREAD:<clientId>,<objectId>,<instanceId>,<resId>

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<objectId>	Integer type Object id number that lwM2M server want to read
<instanceId>	Integer type Instance id number that lwM2M server want to read
<resId>	Integer type Ressource id number that lwM2M server want to read

Example:

+LWM2MREAD:0,3306,111,5750

3.4.6 +LWM2MWRITE Indicator of LwM2M server's writing command

This is an unsolicited message to represent lwM2M client receive the LwM2M server's writing command.

+LWM2MWRITE

+LWM2MWRITE:<clientId>,<objectId>,<instanceId>,<num>[,<resId>,<type>,<length>,<valueStr>]^{num}

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<objectId>	Integer type Object id number that lwM2M server want to write
<instanceId>	Integer type Instance id number that lwM2M server want to write
<num>	Integer type The number of resources need to be write
<resId>	Integer type Resource id number that lwM2M server want to write
<type>	String type "S" String type "O" Opaque type "I" Integer type "F" Float type

<length>	Integer type value length in bytes
<valueStr>	String type value need to write to resource

Example:

+LWM2MWRITE:0,3306,111,5750,O,4,"54595045"

3.4.7 +LWM2MEXECUTE Indicator of LwM2M server's execution command

This is an unsolicited message to represent lwM2M client receive the LwM2M server's execution command.

+LWM2MEXECUTE

+LWM2MEXECUTE:<clientId>,<objectId>,<instanceId>,<resId>,<length>,<valueStr>

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<objectId>	Integer type Object id number that lwM2M server want to execute
<instanceId>	Integer type Instance id number that lwM2M server want to execute
<resId>	Integer type Resource id number that lwM2M server want to execute
<length>	Integer type value length
<valueStr>	String type value of execute command

Example:

+LWM2MEXECUTE:0,3303,0,5605,2,"ok"

3.4.8 +LWM2MOBSERVE Indicator of LwM2M server's observation command

This is an unsolicited message to represent lwM2M client receive the LwM2M server's observation command.

+LWM2MOBSERVE

+LWM2MOBSERVE:<clientId>,<oper>,<objectId>,<instanceId>,<resId>

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<oper>	Integer type 0 Observe object instance 1 Cancel observe
<objectId>	Integer type Object id number that lwM2M server want to observe
<instanceId>	Integer type Instance id number that lwM2M server want to observe
<resId>	Integer type Resource id number that lwM2M server want to observe

Example:

To observe 3306/111/5750:

+LWM2MOBSERVE:0,0,3306,111,5750

Cancel observe 3306/111/5750:

+LWM2MOBSERVE:0,1,3306,111,5750

3.4.9 AT+LWM2MREADCONF

This command response lwM2M server's read command

AT+LWM2MREADCONF

Set Command AT+LWM2MREADCONF=<clientId>,<objectId>,<instanceId>,<resId>,<valuetype>,<valuelen>,<value>	Response OK If there is any error, response: +LWM2M ERROR:<err>
Test Command AT+LWM2MREADCONF=?	Response +LWM2MREADCONF:(list of supported<clientId>),(range of supported<objectId>),(range of supported<instanceId>),(range of supported<resId>),(range of supported<valuetype>),(range of supported<valuelen>), "<value>" OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type	
	LwM2M client's id returned by +LWM2MCREATE	
<objectId>	Integer type	
	Object id number	
<instanceId>	Integer type	
	Instance id number	
<resId>	Integer type	
	Resource id number	
<valuetype>	Integer type	
	0	string
	1	opaque
	2	integer
	3	float
	4	bool
	Other value	undefined
<valuelen>	Integer type	
	Value length	
<value>	String type	
	Value, if type is opaque, value in hex string format	

Example:

Value type is string:

AT+LWM2MREADCONF:0,3306,0,5750,0,5,"hello"

Value type is opaque:

AT+LWM2MREADCONF:0,12001,0,4,1,5,"3432383330"

Value type is integer:

AT+LWM2MREADCONF:0,3306,0,5851,2,3,"206"

Value type is float:

AT+LWM2MREADCONF:0,3303,0,5601,3,4,"3.14"

Value type is bool:

AT+LWM2MREADCONF:0,3306,0,5850,4,1,"1"

3.4.10 AT+LWM2MWRITECONF

This command response lwM2M server's write command

AT+LWM2MWRITECONF

Set Command AT+LWM2MWRITECONF=<clientId>,<result>	Response OK If there is any error, response: +LWM2M ERROR:<err>
Test Command AT+LWM2MWRITECONF=?	Response +LWM2MWRITECONF:(list of

	supported<clientId>),(range of supported<result>) OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<result>	Integer type; result of write command
0x44	Write success
0x8c	Time out
0x84	Object not found
	Refer to rfc 7252

Example:

AT+LWM2MWRITECONF:0,68

3.4.11 AT+LWM2MEXECUTECONF

This command response lwM2M server's execute command

AT+LWM2MEXECUTECONF

Set Command AT+LWM2MEXECUTECONF=<clientId>,<result>	Response OK If there is any error, response: +LWM2M ERROR:<err>
Test Command AT+LWM2MEXECUTECONF=?	Response +LWM2MEXECUTECONF: (list of supported<clientId>) , (range of supported<result>)
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<result>	Integer type; result of write command
0x44	execute success
0x8c	Time out
0x84	Object not found
	Refer to rfc 7252

Example:

AT+LWM2MEXECUTECONF:0,68

3.4.12 AT+LWM2MNOTIFY

Notify LwM2M server a specified resource changed

AT+LWM2MNOTIFY

Set Command AT+LWM2MNOTIFY=<clientId>,<objectId>,<instanceId>,<resId>	Response OK If there is any error, response: +LWM2M ERROR:<err>
Test Command AT+LWM2MNOTIFY=?	Response +LWM2MNOTIFY: (list of supported<clientId>) , (range of supported<objectId>) , (range of supported<instanceId>) , (range of supported<resourceId>) , (range of supported<valuetype>) , (range of supported<valuelen>) , "<value>" OK
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<objectId>	Integer type Object id number
<instanceId>	Integer type Instance id number
<resId>	Integer type Resource id number

Example:

AT+LWM2MNOTIFY:0,3306,111,5851

AT+LWM2MNOTIFY=0,3303,0,5601,3,4,"3.14"

3.4.13 AT+LWM2MUPDATE

This command updates the register information, with or not with object id's update

AT+LWM2MNOTIFY

Set Command AT+LWM2MUPDATE=<clientId>[, <withobj>]	Response OK If there is any error, response: +LWM2M ERROR: <err>
Test Command AT+LWM2MUPDATE=?	Response +LWM2MUPDATE: (list of supported<clientId>), (list of supported<withobj>) OK
Maximum Response Time	65s
Parameter Saving Mode	NO_SAVE

Parameter

<clientId>	Integer type LwM2M client's id returned by +LWM2MCREATE
<withobj>	Integer type 0 not update the object information 1 Update the object information

Example

```
AT+LWM2MUPDATE=0
OK
```

3.4.14 <err>

<err>	Description
ERROR	wrong command name entered, for example, contains Chinese characters
PARAMETER ERROR	parameter error, for example, parameter out of range
CANNOT CREATE SEMPH	can't create semaphore
CONFIG ERROR	configuration lwm2m client error
NO FREE CLIENT	no free client left, now only support one client at one time
OPERATION NO SUPPORT	not support operation, such as GET command
NO FIND CLIENT	not find this client
ADD OBJECT FAILED	fail to add object
NO FIND OBJECT	not find this object id

DELETE OBJECT FAILED	fail to delete the object
NETWORK NOT READY	network not ready, can't use data service
INTERNAL ERROR	Internal error
REGISTER FAILED, BAD REQUEST	Bad request(register)
SERVER REGISTER REJECT	Reject by server when register
REGISTER FAILED, BAD PARAMETER	Bad parameter(register)
REGISTER TIMEOUT	Register timeout
SESSION INVALID	Session invalid
OBJECT ALREADY ADD	Object already add

Note: only strings are currently support, error codes will support in future.

4. Appendix

4.1 Power-on inspection process

- (1) AT // Determine if the module is powered on and started successfully
- (2) AT+CFUN=1 // Turn off airplane mode
- (3) AT+CEREG? // determine the PS domain attachment status. The second parameter is 1 or 5 indicating normal attachment.

4.2 Client login process

- (1) AT+LWM2MCREATE="Leshan.eigencomm.com",5683,56830,"ec-lient-128",86400 // create instance and register
- (2) AT+LWM2MADDOBJ=0,3303,0,4,"5601;5602;5700;5701" // add object with instance id and resource id

4.3 Read/Write/Execute process

- (1) AT+LWM2MREADCONF=0,3303,0,5601,3,4,"3.14" //read command reply
- (2) AT+LWM2MWRITECONF=0,68 //write command reply
- (3) AT+LWM2MEXECUTECONF=0,68 //execute command reply
- (4) AT+LWM2MNOTIFY=0,3303,0,5601,3,4,"3.14" // notify operate
- (5) AT+LWM2MUPDATE=0 // update operate

4.4 Deregister process

- (1) AT+LWM2MDELETE=0 //deregister and client instance destruction

5. Version

Version	Date	Comments
Draft	2018-11-11	
A	2020-01-03	English version
V1.1	2020-3-16	Add some note
V1.2	2020-4-20	Add encrypted client example
V1.3	2020-5-15	Add at command example, change the maximum response time, add introduction about at+lwm2mnotify